

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 6+ years designing highly scalable, highly transactional distributed systems: microservices on Docker and Kubernetes, message queues (Apache Kafka, Google Pub/Sub, NATS), REST APIs, and SQL/MongoDB data layers on AWS and GCP. Led a zero-downtime migration of 20+ production applications, building an idempotent two-way MongoDB↔Firestore sync pipeline that kept two datastores consistent under parallel live traffic. Polyglot engineer with strong object-oriented design across JavaScript/Node.js, Python, and Go — and a proven record of ramping onto unfamiliar stacks fast, going from contractor on a new platform to leading the company-wide migration and writing the SDK documentation other engineers onboarded from. Remote-first since 2020 across US, Singapore, and African teams.

CORE COMPETENCIES

Microservices Architecture

Distributed Systems

Highly Transactional Systems

Docker & Kubernetes

Message Queues (Kafka · Pub/Sub · NATS)

REST API Design

SQL & MongoDB

AWS & GCP Cloud

WORK EXPERIENCE

MTailor Sep 2022 – Present

Software Engineer — US Remote

- Led the end-to-end migration of 20+ production applications from Parse/MongoDB to Firebase/GCP with zero downtime, reporting directly to the CTO.
- Built an idempotent, real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub, keeping both datastores consistent while each served live production traffic — redelivery-safe message processing with conflict handling.
- Implemented Express Shipping end to end on a revenue-critical path, generating an additional \$40 per order.
- Built a 3D visualisation feature using video overlay and ffmpeg, increasing buyer conversion.
- Consolidated infrastructure off AWS (S3/EC2/EBS) onto GCP, saving \$5,000/month; migrated file storage from S3 to GCS via a Python script.

Stack: Node.js, Firebase, GCP Pub/Sub, AWS S3/EC2/EBS, Python, ffmpeg, React, JavaScript

MTailor Mar 2022 – Jul 2022

Contractor (Backend Migration) — US Remote

- Hired onto an unfamiliar stack (Firebase/GCP) and ramped fast: built the MongoDB→Firestore migration scripts with complex processing logic under contract scrutiny.
- Wrote the Firebase SDK documentation (Node.js and Python) that fellow engineers onboarded from; trained the Ops team on dashboards and backup procedures — converted to full-time and given leadership of the entire migration.

Stack: Node.js, Firebase, GCP, AWS, JavaScript

Dr Wealth Aug 2021 – Jan 2022

Software Engineer — Singapore Remote

- Led optimisation of a real-time stock-price PWA: built batched jobs querying 2M+ Firestore records, optimising query shape and read patterns to keep customer-facing prices current from Morningstar APIs at low cost.
- Extended backend REST APIs using Firebase Cloud Functions and Express.js.

Stack: Node.js, Firebase, Express.js, React, JavaScript

Mind2matter Jan 2022 – Jul 2022

Software Engineer — US Remote

- Built backends for DeFi applications using Web3 and Node.js; delivered React UIs from Figma designs under tight agency timelines, reporting directly to the CTO.

Stack: Node.js, Web3, Firebase, React, JavaScript

CodeBits Jan 2020 – Jul 2021

Software Engineer (Team Lead) — Uganda Remote

- Architected an event-driven microservices backend for FIDA Uganda's case management system using Apache Kafka, Docker, and Kubernetes — designed service boundaries, containerised deployment, and led a team of 4 developers through delivery and mentoring.

- Built a USSD service for legal-aid providers communicating with a Node.js backend via gRPC, with a versioned provider-agnostic contract multiple third parties integrated against.
- Built a case management system for FIDA Uganda and a paralegal database for LASPNET, plus a cross-platform React Native mobile app with Firebase push notifications.

Stack: Node.js, Apache Kafka, Docker, Kubernetes, NATS Streaming, gRPC, Firebase, React Native, JavaScript

PROJECTS

Real-time Two-way Sync Pipeline (MTailor)

Message-driven MongoDB↔Firestore synchronisation backbone supporting parallel production traffic on two datastores for months during a phased migration — idempotent processing designed for redelivery from day one, with zero data loss.

Node.js · Google Pub/Sub · MongoDB · Firestore

Kafka/Kubernetes Microservices Platform (FIDA Uganda)

Production event-driven microservices backend serving web, mobile, and USSD clients — event boundaries designed to avoid the distributed-monolith trap, deployed on Kubernetes with Docker.

Apache Kafka · Docker · Kubernetes · gRPC · Node.js

Marketplace Recommendation Engine

Collaborative + content-based filtering engine in MongoDB for service-provider matching, ranking matches by confidence over messy user-generated records in a live marketplace.

MongoDB · Node.js

EDUCATION

BSc Computer Engineering — [Makerere University](#), Kampala

2018 – 2022

SKILLS

Proficient: Node.js, JavaScript/TypeScript, SQL, MongoDB, Firebase, GCP, Python, Git, Linux **Intermediate:** Go, AWS, Rust, React Native

Infrastructure: Docker, Kubernetes, Apache Kafka, Google Pub/Sub, NATS, gRPC **Languages:** English, Kiswahili